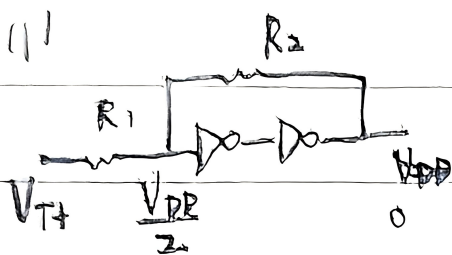


作业13 7.3.4.5.9.10

7.3



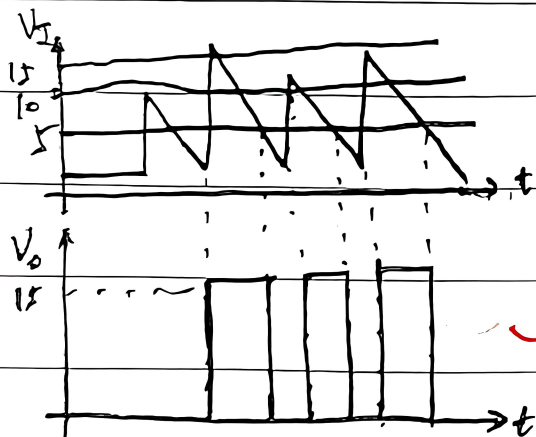
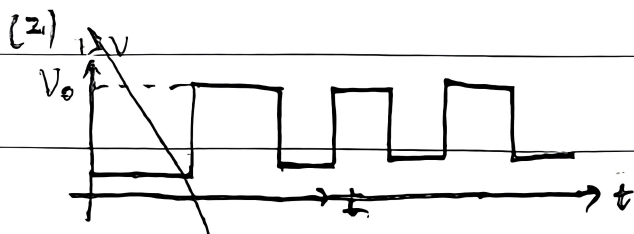
$$\frac{V_{T+} - \frac{V_{DD}}{2}}{R_1} = \frac{\frac{V_{DD}}{2} - V_{DD}}{R_2}$$

$$V_{T+} = 5V$$

$$\frac{V_{T+} - \frac{V_{DD}}{2}}{R_1} = \frac{\frac{V_{DD}}{2}}{R_2} \quad V_{T+} = 10V$$

$$\frac{V_{T-} - \frac{V_{DD}}{2}}{R_1} = \frac{\frac{V_{DD}}{2} - V_{DD}}{R_2} \quad V_{T-} = 5V$$

$$AV_T = 5V$$



7.4

V_{T+} 临界方程: $\frac{V_{T+} - \frac{V_{DD}}{2}}{R_1} + \frac{V_{CO} - \frac{V_{DD}}{2}}{R_3} = \frac{V_{DD}}{2R_2}$

$$V_{T+} = -\frac{R_1}{R_3} V_{CO} + V_{DD} \left(\frac{1}{2} + \frac{R_1}{2R_3} + \frac{R_1}{2R_2} \right)$$

$$\frac{V_T - \frac{V_{DD}}{2}}{R_1} + \frac{V_{CO} - \frac{V_{DD}}{2}}{R_3} = \frac{-\frac{V_{DD}}{2}}{R_2}$$

$$V_{T-} = -\frac{R_1}{R_3} V_{CO} + V_{DD} \left(\frac{1}{2} + \frac{R_1}{2R_3} - \frac{R_1}{2R_2} \right)$$

$$\Delta V_T = \frac{R_1}{R_2} V_{DD}$$

7.5

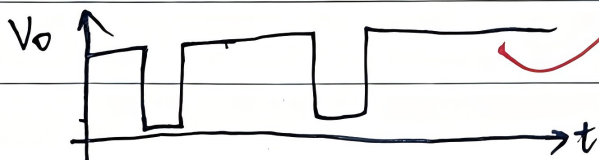
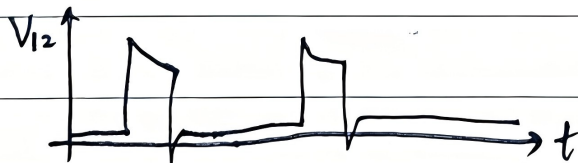
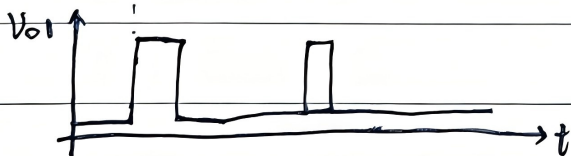
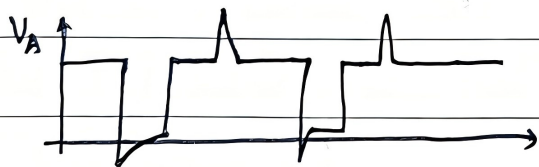
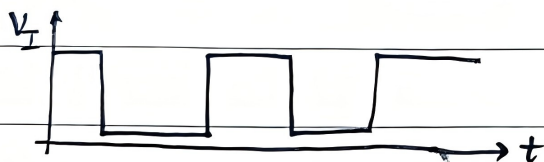
$$\Delta V_T = 2 \frac{R_1}{R_2} \frac{V_{DD}}{2} = \frac{R_1}{R_2} V_{DD}$$

$$\frac{V_{OH} - V_{TH}}{R_2} \leq 4 \text{ mA}$$

$$R_2 \geq 587.5 \Omega$$

$$R_2 = 2.5 R_1$$

7.9



7.10.

$$t_W = R C \ln \frac{V_{DD} - V_{TH}}{V_{TH} - 0}$$

$$= 2.7 \mu\text{s}$$

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